

SIVARAJ MANOHARAN

PROFILE SUMMARY

- Result-oriented professional with **over 18 years** of extensive experience in full-stack development in .net and python, data science, and machine learning across various industries including sales, retail, oil and gas, robotics operations and finance.
- Expertise in developing and deploying machine learning models to optimize pricing strategies, demand forecasting, and customer segmentation, driving strategic decision-making in the retail sector.
- Expertise in identifying & developing innovative and out-of-the-box analytical solutions using suitable machine learning algorithms and statistical methods to achieve actionable insights & optimize business decisions.
- Experience with AWS services, CI/CD pipelines, and big data technologies like Hadoop and Spark, ensuring efficient data processing and deployment.
- Skilled in creating interactive dashboards with Power BI and Tableau, providing real-time insights to business stakeholders for informed decision-making.
- Excels in .NET technologies, with a track record of developing and supporting applications for customer relationship management, and HR systems.
- Capabilities in influencing the strategic direction of the company by identifying opportunities in large, rich data sets and creating and implementing data-driven strategies that fuel growth, including revenue and profits.
- A collaborative leader with exceptional attention to detail, continuously staying updated with the latest industry trends to drive innovation and maintain a competitive edge.

WORK EXPERIENCE

Jul'20 – Oct'24 | Lead Data Scientist(AIML) - ARMADIO LLC, Abudhabi

Key Result Areas:

- Leveraging advanced AI research methodologies, statistical methods, and machine learning algorithms using python to analyze extensive Sales, retail, Robotics Operations, finance datasets, generating actionable insights that drive strategic decision-making and improve business outcomes.
- Developing and refining predictive models for optimizing pricing strategies, demand forecasting, inventory management, and customer segmentation to boost operational efficiency.
- Collaborating with business stakeholders to pinpoint key challenges and opportunities in the retail sector, finance and Sales, transforming these insights into effective, data-driven solutions.
- Designing and implementing scalable machine learning pipelines, covering data preprocessing, feature engineering, model training, and deployment to enhance operational workflows.
- Conducting comprehensive model evaluations, including A/B testing, and iteratively improve model performance to ensure accuracy and effectiveness.
- Analyzing and forecasting Robotics and sensors logs data using advanced neural networks and state-of-the-art AI technologies such as NLP, RAG, and BERT models.
- Analyzing and forecasting sales, retail, robotics operations and financial data using GenAI LLM models like LLAMA3, GPT4, advanced neural networks and state-of-the-art AI technologies such as NLP, RAG, and BERT models.
- Experience with AWS and Google Cloud Platform, including model deployment, data pipeline creation, and MLOps. Familiar with cloud-based machine learning services and able to adapt to Azure ML and Databricks

Highlights:

- Developed predictive models that optimized pricing strategies, demand forecasting, inventory management, and customer segmentation, leading to a 20% increase in sales and a 30% improvement in forecasting accuracy.
- Implemented advanced AIML and analytical techniques that reduced inventory costs by 15% while enhancing overall operational efficiency and profitability.

Targeting opportunities: Lead Data Scientist, Lead AI ML Specialist, Lead Machine Learning Expert, and Lead Business Intelligence Specialist leveraging advanced analytics to drive strategic decision-making, optimize processes, and enhance customer engagement.

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EDUCATION

- Bachelor of Engineering (Computer Science) from Sri Subramanya College of Engg. and Tech in 2004

CORE COMPETENCIES

- Big Data Analytics
- Machine Learning
- Artificial Intelligence
- Deep Learning
- NLP (Natural Language Processing)
- Model Training and Deployment
- Data Preprocessing and Feature Engineering
- Agile Methodologies
- Model Evaluation and A/B Testing
- Scalable Data Processing
- Data Visualization
- Neural Networks
- Statistics
- Data Governance

TECHNICAL SKILLS

- **Programming Languages and Technologies:** Python | Shell Scripting | JavaScript | ASP.NET | C#.NET
- **Data Science and Machine Learning:** Pytorch | Tensorflow | XGBoost | Computer Vision | RAG Model | BERT | ChatGPT-4 | Hugging Face
- **Data Storage and Databases:** MS SQL Server | MongoDB
- **Big Data Technologies:** Spark | Hadoop
- **Cloud Services:** AWS S3 | AWS Lambda | Azure
- **Data Visualization:** Tableau | PowerBI
- **DevOps and CI/CD:** MLOps | Docker & Kubernetes
- **Frameworks:** Keras

SOFT SKILLS

- Time Management
- Analytical
- Communicator
- Problem-solving
- Team Management

CERTIFICATIONS

- ITIL Foundation V3
- PMP - Project Management Professional
- CSM - Certified Scrum Master

PERSONAL DETAILS

- **Address:** Chennai, Tamilnadu, India

- Led AI research initiatives that analyzed and forecasted market trends, driving strategic insights and innovative solutions for data-driven business transformations.
- Implemented data governance frameworks ensuring compliance with GDPR and other data protection regulations. Led initiatives to mitigate cybersecurity risks in machine learning pipelines.

Jun'17 - Jun'20 | Senior Consultant | Okool, Abu Dhabi, UAE

Key Result Areas:

- Led a team of data Scientist, Data Engineers in optimizing oil and gas operations by implementing data-driven strategies.
- Created machine learning models for predictive maintenance and anomaly detection, leading to significant improvements in operational efficiency.
- Leveraged AWS services to handle large-scale data processing, ensuring both scalability and reliability of data operations.
- Implemented CI/CD pipelines to enhance the efficiency of continuous integration and deployment processes.
- Applied NLP-based analysis using advanced LLM techniques to gain deeper insights from complex data sets.
- Analyzed and forecasted oil and gas drilling sensor data, machine logs, and geological data using sophisticated neural networks and AI technologies in python, including NLP, RAG, and BERT models.
- Developing and implementing global and functional performance management scorecards.
- Championing data governance and insights-driven decision-making across the organization.
- Managing and mentoring a global team, fostering a culture of continuous learning.

Highlights:

- Ensured solutions adhered to business rules and industry standards, which improved accuracy in anomaly detection by 50% and cut maintenance activities by 50%.
- Proficient in MLOps practices, including continuous integration and deployment (CI/CD) using Docker, Jenkins, and GitHub Actions. Successfully automated deployment pipelines, reducing model update times by 30%.
- Ensured adherence to ISO 27001 standards in data handling and storage processes. Conducted regular compliance audits to maintain data integrity and security.

Jun'16 - Feb'17 | Business Consultant | Tech Mahindra, Melbourne, Australia

Key Result Areas:

- Analyzed and predicted customer data to enhance profitability and drive improved business outcomes; developed predictive models to estimate customer rebooking likelihood, aiding in targeted marketing initiatives.
- Employed big data analytics to devise strategic business solutions and optimize operational efficiency.
- Utilized Power BI and python for effective data visualization, providing clear and actionable insights to support informed decision-making.
- Forecasted sales and logistics data for a manufacturing company using advanced machine learning and deep learning models.

Highlight:

- Successfully established a data-driven culture leading to enhanced decision-making across the organization.
- Leveraged advanced analytics to deliver strategic business solutions, resulting in a 25% reduction in expenses and a 15% increase in sales.

Mar'12 - Apr'16 | Senior Associate | Cognizant Technology Solution, Chennai

Key Result Areas:

- Developed advanced ML algorithms in python for banking and financial stock management, including predictive models for risk assessment, fraud detection, and portfolio optimization, leading to enhanced operational efficiency and improved decision-making.
- Designed and implemented ML-driven solutions to streamline healthcare and insurance processes, including claims processing automation and customer

segmentation, resulting in improved workflow efficiency and service delivery.

- Created and deployed machine learning models for predictive maintenance and anomaly detection in python, utilizing techniques like time-series analysis and unsupervised learning to identify potential issues proactively, significantly improving operational efficiency and reducing downtime.
- Built and integrated robust testing frameworks to validate the performance, accuracy, and scalability of ML algorithms, ensuring high reliability and stability of applications across diverse environments.
- Simplified patient healthcare information management through NLP and classification models, enhancing data organization, accessibility, and handling for both patients and providers.

Highlight:

- Developed robust analytics frameworks that significantly improved reporting accuracy and efficiency.
- Streamlined the process for managing patient healthcare information, which reduced lab response time by 25%.

Nov'09 - Feb'12 | Module Lead | Mphasis, Chennai

Key Result Areas:

- Designed web services and maintained detailed architecture documentation to facilitate development and support.
- Implemented automated testing protocols to enhance code coverage and ensure high software quality.
- Developed and maintained applications for customer relationship and credit information management, collaborating closely with clients to meet evolving requirements using Agile methodologies.

Highlight:

- Led a team of software engineers, achieved a 40% reduction in development time through effective project management and mentoring.

Sep'07 - Nov'09 | Senior Software Engineer | Hexaware Technologies

Key Result Areas:

- Developed HR and stock management systems to boost organizational efficiency and streamline processes.
- Designed and implemented web and desktop applications using .NET technologies and MSSQL Server to enhance system functionality.
- Automated bulk data uploading and stock allocation processes, significantly improving operational efficiency.

Mar'06 - Aug'07 | Software Engineer | Intek Systems PVT LTD,Maldives

Key Result Areas:

- I have a track record of developing automated scripts that streamline routine tasks, significantly reducing manual effort and enhancing operational efficiency.
- I am proficient in developing REST- APIs and implementing robust authentication protocols like OAuth to securely expose data to clients while prioritizing data security.
- In my role, I have prepared web applications for presentation to clients, ensuring they meet high-quality standards and are ready for client-facing interactions.
- I am dedicated to maintaining quality control through activities such as integration testing and conducting thorough code reviews to ensure code quality and reliability.